LESSON PLAN

HELICOPTER THEORY (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	16
Total No. of Lesson Plans	27	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Secondary/Auxiliary flight controls, thrust reversing/braking
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Secondary/Auxiliary flight controls, thrust reversing/braking
5. Specific Objectives	11.1.2.1 Trainee will be able to describe various types of aerofoil sections, conditions of performance, theories, and concept related with performance

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of Secondary Control Systems diagrams; Safety: Do not operate actual flight controls without instructor supervision; Instructor supervision required for all practical discussions
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary flight controls?
(b) Define the function of an auxiliary flight control.
(c) List two conditions affecting aerofoil performance.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Secondary Flight Controls • Types of Spoilers and Flaps • Function of Trim Systems • Aerodynamic Theory of Control Surfaces • Performance Conditions for Aux Controls	L & Ds	PPT, IB	15 min
(b)	• Thrust Reversing Mechanisms • Braking Systems Operation • Hydraulic Power for Reversers • Safety Interlocks in Reversers • Maintenance Checks for Brakes	L & Ds	PPT, PDF	20 min
(c)	• Integration of Aux Controls with Primary • Effect of Spoilers on Lift Distribution • Brake Fade and Thermal Limits • Reverser Deployment Sequence • Troubleshooting Aux Control Failures	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how thrust reversers assist in landing rollout.
(b) Describe the role of trim tabs in secondary control systems.
(c) Identify three safety interlocks used in thrust reversing systems.

12. Home Assignment and Follow Up	Prepare a brief note on the differences between primary and secondary flight controls, referencing the concepts discussed in LP 17 (Flight control surface balancing).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Flight control surface balancing. Flow Separation Stalling, behaviour of aircraft before and after stall. Factors effecting stalling speed, high incidences stall, high speed stall, G stall, wing tip stall, Basic stalling speed.	Refer to the master syllabus reference list	LP 17

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Engineer, P No. 12345	Senior Instructor, Chief Engineer, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.
6. Verify specific brake thermal limit values for current helicopter fleet.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min